

PowerPanel™ Systems Installation and Operations Manual

2026, By Guthrie Hausken for OSU TW-SSM.

Introduction

This manual will outline the proper operation, safety procedures, and maintenance procedures for the PowerPanel™ fuel dispensing system. It provides information for qualified technicians responsible for the installation, start-up, operation, and maintenance of fuel dispensing equipment. This manual also covers compliance with industry regulation, and safe system operation guidelines.

Product owners should reference this manual for safety information, emergency shutdown procedures, Lockout/Tagout protocols, system commissioning, and ongoing maintenance. This document covers warning label classifications, regulatory compliance requirements, operational procedures under normal and emergency conditions, and recommended maintenance intervals.

2.0 Installation and Start-Up

This section describes the general installation of PowerPanel™ systems. Refer to site-specific documentation for detailed wiring and equipment hookup instructions.

2.1 Installation of PowerPanel™ Integrated Systems

All on-site preparation should be completed prior to delivery and installation of any PowerPanel™ system.

2.1.1 Site Template

If you need assistance placing and arranging the conduit stub-ups prior to the concrete slab pour, our engineering team can send a site template. The template will be site-specific, and while not required, it is recommended to ensure proper installation of the equipment.

If you are unsure whether or not you will receive this template, please contact our engineering staff.

Chase drawings are also available after the job is ordered.

Contact engineering for more information.

2.1.2 Equipment Delivery and Acceptance

On the day of equipment delivery, before you sign off on the receipt, closely inspect all shipping skids, banding, etc. to ensure there is no damage from transit. We provide an equipment list to verify that all required equipment was shipped.

Any discrepancies or damage must be noted on the delivery ticket and have the driver's initials, indicating it as having been noted.

2.1.3 Setting up the Equipment

Most of the equipment we ship will be on a shipping skid that can be removed by a heavy-duty forklift. In some cases, the equipment will be delivered on an open flat-bed trailer and can be removed with a crane and lifting hoist.

Once the equipment is permanently in place, the lifting bars mounted at the top of the unit can be removed. Please refer to the *Installation Lifting Drawing* for special sealing instructions regarding the NEMA 3R design. There are seals included with the equipment that must be installed before a unit can be approved by inspection.

Refer to the provided documentation to proceed with the rest of equipment installation. The first item to be completed is the permanent securing of the PowerPanel™ to the concrete mounting pad.

Please check the sizes of all main breakers, lugs, and feed wires for compatibility before proceeding.

3.0 System Start-Up and Maintenance

This section describes standard lockout procedures, Initial start-up and commissioning, operational procedures, and routine maintenance of PowerPanel™ systems.

3.1 Lockouts

Electrical contractors or anyone qualified to work on the PowerPanel™ must be familiar with proper lockout procedures.

The breakers for this panel are capable of being locked out during maintenance operations or, if otherwise required, of being locked in the ON position.



Figure 3.1-1 Typical Circuit Breaker and Breaker Lockout/Lockon Adaptor

Critical or life safety loads require the power source to be locked in the ON position so there will be no accidental disruption of service.

3.1.1 Lockout Procedure

Follow this procedure to lock out a breaker.

1. Turn designated breaker to the OFF position.
2. With your thumb and forefinger, squeeze the retainer clips of the lockout together. Locate the two retainer holes just below the breaker handle.

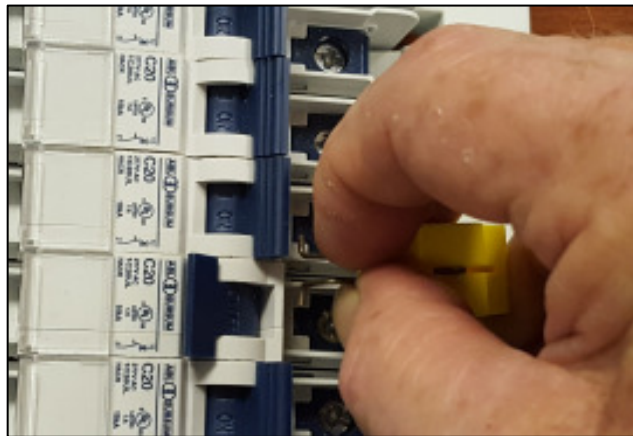


Figure 3.1.1-1 Applying Lockout

3. Slip the retainer pins into the two holes and release the clips.



Figure 3.1.1-2 Lockout Attached

4. Raise the tab of the lockout and push down towards the retaining clips. The body will slide down over the clips, and the lockout loop will appear near the top.

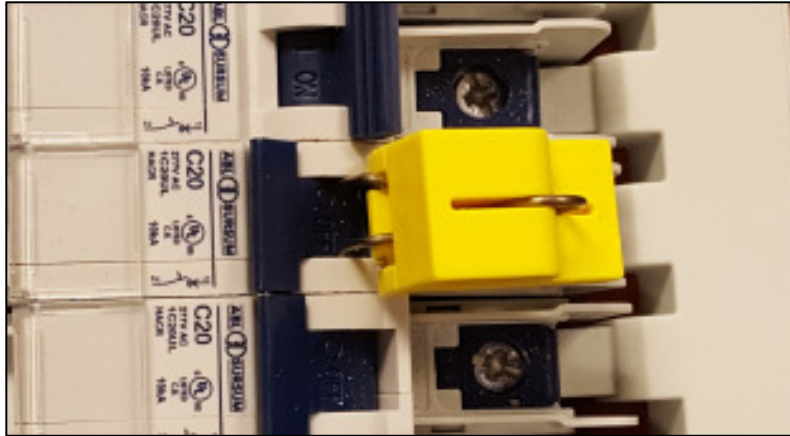


Figure 3.1.1-3 Lockout in Place

5. With the lockout loop exposed, attach a padlock. The lockout will accommodate a lock with approximately a 5/16" shank. If you have a TAG-OUT procedure in place, apply notice at this time.



Figure 3.1.1-4 Breaker Locked Out

If you are unsure or have questions regarding this or any other procedure described in this manual, please contact your local representative.

3.1.2 Other Lockouts

The main breaker that feeds power to the PowerPanel™ needs to be locked out if servicing the PowerPanel™ is required.

Consult the breaker manufacturer's procedure to perform the lockout.

If your unit includes the optional STP breaker and bus assembly, use the lockout procedure as listed in 3.1. The lockouts will be different since they are made to accommodate 2 or 3 pole breakers.

The optional assembly has a separate feed breaker powering the bus assembly. Consult the manufacturer's recommendations for the proper lockout method.

3.4 Maintenance

The PowerPanel™ needs minimal maintenance to keep it in good working order.

3.4.1 Monthly

1. Perform an Emergency Fuel Shutdown and Reset operation to verify that the system shuts down and resets correctly. If there are multiple shutdowns, make sure each shutdown will turn the system OFF.
2. Open the Fueling Panel section and inner door. Visually inspect the operational lights on the TVSS to verify that it is in operational mode.
3. For NEMA 3R enclosures, verify that the 3-point door latches are adjusted properly. When adjusted properly, the enclosure door will compress about half of the door gasket's perimeter.

3.4.2 Semi-Annually

1. Inspect all door gaskets for any damage. Replace any damaged gaskets immediately.
2. Inspect interior of cabinets (Both LV Section & Fuel Panel sections) for any buildup of moisture or other aggregates. Clean as required.
3. Check the incoming voltage at the panelboard main breaker for all 3 phases and neutral.

3.4.3 Annually

1. Power down the main breaker of the panelboard. Verify that all wiring connections are tightened properly.
2. Perform random 'pull' tests on some of the wiring to verify that no loose wiring exists.